



SPRFMO

South Pacific Regional Fisheries Management Organisation

Jack Mackerel Working Group

Benchmark meeting report 2026

May 18, 2026

Summary

This is a starter draft for the 2026 Jack Mackerel Working Group benchmark meeting report. It is structured to capture the main workshop decisions, agreed recommendations, unresolved technical issues, and follow-up tasks needed to support the June 2026 management strategy evaluation workshop and subsequent Scientific Committee review.

The benchmark meeting is expected to focus on assessment inputs, abundance-index evaluation, biological data and assumptions, assessment-model simplification, operating-model conditioning, and priorities for the MSE work programme. This draft should be updated during the workshop as sessions conclude, with short decision text and action items added while technical discussions are still current.

Meeting Details

- Meeting: JMWG benchmark meeting 2026
- Date: 18-22 May 2026
- Location: Lima, Peru
- Organisation: South Pacific Regional Fisheries Management Organisation
- Working group: Jack Mackerel Working Group
- Host: Peru / IMARPE
- Chair: Jim Ianelli
- Meeting format: Primarily in person, with selected remote participation as arranged

Opening of the Meeting

The Chair opened the meeting and welcomed participants to the 2026 benchmark workshop. The Chair noted that the workshop has a practical purpose: to convert a large set of technical contributions into clear benchmark priorities, model-input decisions, and follow-up tasks for the JMWG and MSE process.

The Chair emphasized that the benchmark material contains many sensitivities, including stock structure, index selection, catchability and availability, effort creep, selectivity, recruitment assumptions, biological inputs, and operating-model conditioning. The workshop should focus on which sensitivities materially affect advice, operating-model conditioning, and MSE readiness.

Members were encouraged, to the extent practical, to supplement technical presentations with concise information on data extent, including years covered, spatial coverage, fleet or survey coverage, observation unit, sample sizes, treatment of missing data, and whether uncertainty is available for assessment use.

The Chair also noted that work sessions and report-writing time are part of the technical work of the meeting. Each session should aim to leave behind short text on decisions, unresolved issues, data requests, and recommended follow-up.

Adoption of Agenda

The meeting reviewed the draft agenda and agreed to organize the week around acoustic indices, CPUE indices, biological inputs, assessment-model development, projections, operating-model specifications, and final report adoption. A shortened agenda is provided in Appendix A.

Workshop Objectives

The main objectives of the benchmark meeting are to:

1. Review and document candidate abundance indices and their suitability for assessment use, operating-model conditioning, and possible management-procedure input.
2. Review biological inputs, including maturity, natural mortality, weight-at-age, length-weight relationships, and length-frequency information.
3. Review the SC13 assessment base model and candidate simplifications for benchmark and MSE use.
4. Identify a defensible set of assessment-model alternatives, sensitivity runs, and diagnostics.
5. Define operating-model conditioning priorities and uncertainty dimensions for the June 2026 MSE workshop.

6. Document decisions, unresolved technical issues, and follow-up responsibilities in the meeting report.

Organization of Work

The meeting work should be organized around technical sessions and smaller drafting or analysis groups. Where possible, each major topic should produce:

- a short decision statement;
- a list of accepted inputs or candidate inputs;
- a list of sensitivities or diagnostics to run;
- data or documentation requests;
- text that can be inserted directly into the final workshop report.

For contributors who are not working directly in GitHub, use Google Docs as the collaborative drafting layer and keep this Quarto report as the source of record. The recommended workflow is described in the [Report Collaboration Workflow](#).

Key Discussion Summary

Acoustic Indices

The Day 1 acoustic session combined formal SCW16 working papers with presentation-only material. The presentations covered the history of the north-central Chile acoustic survey programme, preliminary 2026 survey results, oceanographic conditions during the 2026 survey, spatio-temporal standardization of acoustic backscatter, fishery-dependent acoustic information, Peruvian acoustic assessment information, and sonar applications relevant to jack mackerel and anchoveta fisheries.

For reporting purposes, the presentation material should be separated into four categories: formal papers tied to numbered SCW16 documents, presentation-only background, new 2026 evidence, and context or supporting methods. This distinction is important because not all Day 1 material is intended to become a direct assessment input. Some presentations provide candidate abundance indices, while others provide survey context, environmental interpretation, diagnostic information, or methodological background.

A Day 1 presentation crosswalk is provided in Appendix D. That table should be updated as presenters confirm titles, authors, and whether each presentation should be cited as a working paper, supporting presentation, or meeting-record item.

CPUE Indices

Text to be added during the meeting. This section should summarize the CPUE coordination discussion, historical CPUE synthesis, effort-creep treatment, spatio-temporal CPUE alternatives, Peruvian CPUE standardization, index-weighting questions, and whether separate or combined CPUE products should be carried forward.

Biological Inputs

Text to be added during the meeting. This section should summarize biological assumptions and data inputs, including maturity, natural mortality, weight-at-age, length-weight relationships, length-frequency composition data, and implications for assessment and operating-model conditioning.

Assessment Model Development

Text to be added during the meeting. This section should summarize the SC13 base-model developments, candidate simplifications, single-stock and two-stock alternatives, reduced-index configurations, selectivity assumptions, recruitment scenarios, and model diagnostics.

Operating Models and MSE Linkages

Text to be added during the meeting. This section should summarize operating-model specifications, reference-set and robustness-set uncertainties, MCMC feasibility, management-procedure input diagnostics, and workflow needs for the June 2026 MSE workshop.

Decisions and Agreements

Add agreed decisions here as the meeting proceeds.

No.	Decision or agreement	Topic	Follow-up needed
1	TBD	TBD	TBD

Action Items

Add actions here as the meeting proceeds.

Action	Lead	Support	Target timing
Update the participant list and affiliations	Chair / Secretariat	Members	During meeting
Maintain session-level report text as discussions conclude	Session leads	Rapporteurs / participants	Daily
Prepare a final index-ranking table for candidate abundance indices	Assessment/index subgroup	Acoustic and CPUE contributors	Before report adoption
Prepare an operating-model uncertainty table for MSE conditioning	MSE subgroup	Assessment subgroup	Before report adoption

Open Issues / Risks to Track

- Some candidate indices may have useful contrast but unresolved representativeness, catchability, availability, or uncertainty issues.
- Acoustic and CPUE information may give conflicting recent signals, requiring clear documentation of interpretation and model-treatment choices.
- Model simplification for MSE tractability must be balanced against structural assumptions that members view as scientifically important.
- Biological inputs may differ by area, fleet, season, or stock hypothesis, and may require explicit sensitivity tests.
- Report writing must keep pace with the technical discussion so that decisions are not lost before adoption.

Appendix A. Shortened Draft Agenda

This appendix summarizes the draft agenda from SCW16-Doc00. It should be updated if the adopted meeting agenda differs materially.

Day 1: Acoustic Session

- Opening ceremony, official photo, and meeting organization.
- Historical review of hydroacoustic survey programs in north-central Chile.

- Preliminary results of the 2026 north-central Chile hydroacoustic survey.
- Oceanographic conditions during the 2026 acoustic survey.
- Spatio-temporal modeling of acoustic backscatter from the north-central Chile survey.
- Toward a standardized northern Chile acoustic abundance index.
- Opportunistic acoustic data collection from the south-central Chilean commercial fleet.
- Review of jack mackerel acoustic assessments conducted in Peru, including SCW16-Doc16.
- Echogram examination and validation.
- General discussion and technical recommendations.

Day 2: CPUE Session

- Metadata by fleet and country.
- CPUE methods by fleet and country.
- Peruvian CPUE report, including SCW16-Doc15.
- Comparative analysis and parsimony across candidate indices.
- Effort-creep review and decision for stock assessment use.
- Catchability and availability blocks by abundance index.
- CPUE weighting factors for stock assessment.
- Separate versus combined index products for HCR or management-procedure use.

Day 3: Recap, Report Drafting, and Biological Inputs

- Recap of acoustic and CPUE sessions.
- Drafting of index-session report text and follow-up tasks.
- Summary of SCW16-Doc01 and benchmark-model implications.
- Review of biological assumptions used in the assessment and operating models.
- Review of Peru length-weight and length-frequency information, including SCW16-Doc13 and SCW16-Doc14.

Day 4: Projections, Recruitment Scenarios, and Operating Models

- Assessment projections and recruitment scenarios.
- Candidate assessment-model runs and sensitivity structure.
- Operating-model specifications for MSE conditioning.
- MCMC evaluations, convergence diagnostics, run-time constraints, and feasibility.

Day 5: Final Decisions and Report Adoption

- Final benchmark decisions and recommendations.
- Confirmation of model-run priorities, operating-model specifications, and follow-up tasks.

- Report drafting, review, and adoption.

Appendix B. Participants

This participant list is based on the registration information available at the start of the benchmark meeting. Email addresses are retained outside this public report.

Name	Delegation	Role	Organisation / agency	Format
Jose Ignacio Zenteno Loredo	Chile	Advisor	IFOP	In person
Aquiles Sepulveda	Chile	Advisor	INPESCA	In person
Qi Lee	Invited Expert	Invited Expert by SPRFMO	University of Washington	In person
Niels Teon Hintzen	European Union	Expert	PFA	In person
Karolina Molla Gazi	European Union	Designated Representative / HoD	Wageningen Marine Research	In person
Jim Ianelli	JMWG Chairperson	Expert / Chair	NOAA	In person
Victor Catasti	Chile	Expert	IFOP	In person
Ignacio Paya	Chile	Expert	IFOP	In person
Sebastian Vazquez	Chile	Expert	INPESCA	In person
Nicolas Alegria	Chile	Expert	INPESCA	In person
Nicole Mermoud	Chile	Designated Representative / HoD	SUBPESCA	In person
Ana Alegre Norza Sior	Peru	Designated Representative / HoD	Instituto del Mar del Peru	In person
Criscely Lujan Paredes	Peru	Expert	Instituto del Mar del Peru	In person
Mirian Geronimo	Peru	Expert	Instituto del Mar del Peru	In person
Aparicio Gersson Roman Amancio	Peru	Expert	Instituto del Mar del Peru	In person

Name	Delegation	Role	Organisation / agency	Format
Erich Diaz Acuna	Peru	Expert	Instituto del Mar del Peru	In person
Josymar Torrejon	Peru	Expert	Instituto del Mar del Peru	In person
Daniel Isaias Grados Paredes	Peru	Expert	Instituto del Mar del Peru	In person
Carlos Valdez Mego	Peru	Expert	Instituto del Mar del Peru	In person
Marissela Pozada Herrera	Peru	Expert	Instituto del Mar del Peru	In person
Mariano Sergio Gutierrez Torero	Peru	Expert	Instituto Humboldt	In person
Salvador Peraltilla Neyra	Peru	Expert	Sociedad Nacional de Pesqueria	In person
Sandra Marisol Cahuin Villanueva	Peru	Expert	Instituto del Mar del Peru	Virtual
Iago Mosqueira	Expert	Expert	WUR	Virtual
Teobaldo Dioses	Peru	Expert	GMAIL	In person
Jaime Letelier	Chile	Expert	IFOP	TBD
Esteban Molina	Chile	Expert	IFOP	TBD
Francisco Leiva	Chile	Expert	IFOP	TBD
Javier Legua	Chile	Expert	IFOP	TBD
Jorge Castillo	Chile	Expert	IFOP	TBD

Appendix C. Document Summary by Topic

This appendix summarizes SCW16-Doc01 through SCW16-Doc16. The organization is by topic rather than document number so that related evidence can be reviewed together.

Assessment and Model Structure

Developments of the Base SC13 Model

[SCW16-Doc01](#) documents benchmark development from the SC13 JJM assessment model, including single-stock and two-stock configurations, reduced-index sensitivities, selectivity

alternatives, and model diagnostics. It is the main bridge between the current assessment model and the benchmark model set.

Key report use: summarize accepted base-model structure, candidate simplifications, retained sensitivities, and any model configurations to carry into OM conditioning.

Operating Models and MSE

Operating Models for Jack Mackerel MSE

[SCW16-Doc04](#) describes the operating-model structure for jack mackerel MSE. It should be used to organize reference-set and robustness-set uncertainties, including stock hypotheses, recruitment assumptions, selectivity, catchability, and future projection structure.

Key report use: define the operating-model grid or equivalent uncertainty table that will feed the June 2026 MSE workshop.

CPUE

CPUE Coordination Priorities and Working-Paper Plan

[SCW16-Doc02](#) provides the forward-looking CPUE coordination plan for the benchmark, including candidate topics, decision points, metadata needs, and working-paper expectations.

Key report use: structure the CPUE session and make sure each candidate CPUE product is classified for assessment, OM conditioning, MP input, diagnostic use, or hold.

Meta-Analysis of CPUE Papers on Jack Mackerel

[SCW16-Doc03](#) synthesizes CPUE papers from 2022-2025. It provides historical context on CPUE methods, index conflicts, effort creep, spatial behavior, environmental effects, and recurring recommendations.

Key report use: provide background for interpreting new CPUE papers and avoid repeating previously identified unresolved issues.

Chile CPUE Effort Creep

SCW16-Doc07 focuses on implementation of an informed creep correction for the Chilean jack mackerel CPUE index. It is central to decisions about fishing-power correction and whether a revised CPUE series should be used directly or treated as a sensitivity.

Key report use: document whether effort-creep treatment is accepted, treated as a sensitivity, or deferred pending additional diagnostics.

Chile CPUE with sdmTMB/SPDE

SCW16-Doc10 updates the Chilean CPUE abundance index using spatio-temporal SPDE-based models with sdmTMB. It should be compared directly with other Chilean CPUE standardization approaches.

Key report use: evaluate whether the spatio-temporal CPUE product should replace, supplement, or diagnose the current assessment CPUE series.

Chile CPUE with INLA

SCW16-Doc12 standardizes the central-southern Chile CPUE fishery using hierarchical Bayesian models with INLA. It provides a complementary spatio-temporal approach to Doc10.

Key report use: compare diagnostics, uncertainty, spatial extrapolation, and annual index behavior against the sdmTMB CPUE product.

Peru CPUE

SCW16-Doc15 presents standardization of catch-per-unit-effort for jack mackerel from 2015-2025 in Peruvian national jurisdictional waters.

Key report use: evaluate whether the Peruvian CPUE product is suitable as a far-north abundance index, an assessment sensitivity, an OM-conditioning input, or a diagnostic series.

Acoustic Indices

Northern Chile Acoustic Spatio-Temporal Model

SCW16-Doc06 develops a standardized abundance index for Chilean jack mackerel using spatio-temporal modeling of acoustic survey data in northern Chile.

Key report use: evaluate acoustic index representativeness, uncertainty, spatial coverage, and suitability for assessment and OM use.

Fishery-Dependent Acoustics

SCW16-Doc08 summarizes interannual variability in distribution, size structure, and biomass estimated from fishery-dependent acoustics.

Key report use: decide whether fishery-dependent acoustic products provide a usable abundance signal or primarily diagnostic/context information.

Central-North Acoustic Density with sdmTMB

SCW16-Doc09 estimates a jack mackerel acoustic density index for central-north acoustic surveys using spatio-temporal models with sdmTMB.

Key report use: compare with Doc06 and other acoustic products to determine whether indices are complementary, redundant, or conflicting.

Peru Acoustic Index Considerations

SCW16-Doc16 summarizes considerations on the use of the jack mackerel acoustic index in Peruvian national jurisdictional waters.

Key report use: classify the Peru acoustic information as a candidate index, supporting interpretation, or diagnostic evidence, depending on data extent, uncertainty, and assessment compatibility.

Biology and Composition Inputs

Biology

SCW16-Doc05 summarizes biological inputs for the assessment, including natural mortality, maturity, weight-at-age, and spawning-biomass calculations.

Key report use: document accepted biological assumptions and identify sensitivity tests that should be carried into assessment or OM conditioning.

Peru Length-Weight Relationship

SCW16-Doc13 presents the length-weight relationship of jack mackerel in Peru and implications for biomass estimation.

Key report use: evaluate whether Peru-specific length-weight information affects biomass conversion, biological assumptions, or regional differences in assessment inputs.

Peru Length-Frequency Update

[SCW16-Doc14](#) summarizes updated length-frequency data for the Peruvian jack mackerel far-north stock in Peruvian jurisdictional waters.

Key report use: document the composition-data evidence available for the far-north stock and decide how it should be used in assessment and stock-structure discussions.

Data Quality and Supporting Evidence

Catch/Fishery Data Quality Checks

[SCW16-Doc11](#) provides catch and fishery data quality-control checks.

Key report use: use as a support screen for catch- and fishery-dependent indices, and document any data-quality issues that affect benchmark decisions.

Appendix D. Day 1 Presentations

This appendix records the Day 1 presentation files available in `docs/day1`. It is intended as a crosswalk between the acoustic-session agenda, formal SCW16 papers, presentation-only material, and how each item should be used in the meeting report.

Presentation topic	Linked paper	Main contribution	Report use
Historical review of hydroacoustic survey programs in north-central Chile	Presentation only	Survey-program history, continuity, and context for interpreting the northern Chile acoustic time series.	Background / context
Preliminary results of the 2026 north-central Chile hydroacoustic survey	Presentation only; related to Chile acoustic index papers	New 2026 acoustic information and immediate interpretation of recent survey conditions and abundance signal.	New 2026 evidence / benchmark caveat

Presentation topic	Linked paper	Main contribution	Report use
Oceanographic conditions during the 2026 acoustic survey	Presentation only	Environmental context for interpreting the 2026 survey and potential availability or distribution effects.	Context / supporting interpretation
Spatio-temporal modeling of acoustic backscatter using sdmTMB	SCW16-Doc09	Acoustic density index estimated with spatio-temporal methods.	Formal paper / candidate index
Standardized northern Chile acoustic abundance index	SCW16-Doc06	Spatio-temporal modelling of northern Chile acoustic-survey data.	Formal paper / candidate index
Opportunistic acoustic data collection using the south-central Chilean commercial fleet	SCW16-Doc08	Fishery-dependent acoustic information on distribution, size structure, and biomass.	Formal paper / diagnostic or candidate index
Review of jack mackerel acoustic assessments conducted in Peru	SCW16-Doc16	Peruvian acoustic assessment context and considerations for index use.	Formal paper / supporting evidence or candidate index
Sonar applications in Peruvian fisheries and applicability for jack mackerel	Presentation only	Methods and operational context for sonar/acoustic information in Peru and Chile.	Context / supporting methods

Use the following labels when carrying presentation material into the report:

Label	Meaning
Formal paper	Presentation tied to a numbered SCW16 working paper.
Presentation only	Material presented to the meeting but not currently represented by a numbered SCW16 paper.
New 2026 evidence	Recent information that may affect benchmark interpretation even if it is not a standalone paper.

Label	Meaning
Context / supporting methods	Background or method information that helps interpret candidate indices but is not itself a direct assessment input.
